MD. Jawadul Hasan

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Profile

Software engineer with three years of experience building enterprise systems, backend services, and AI applications. Passionate about developing scalable, intelligent systems. Currently exploring how to make AI agent outputs more trustworthy by tracing answers to source evidence and distinguishing computed results from model-generated inferences.

Education

Åbo Akademi University, Turku, Finland August 2026 – Present

M.S. in Computer Science, Erasmus Mundus Joint Master's Programme (EDISS)

- Research topic: Building a skepticism layer for LLM-based agentic systems that shows where each answer comes from, separates results computed from source data from model-generated inferences, and surfaces supporting evidence, so users can spot hallucinated claims and decide when to trust the system.
- Relevant coursework: Data Science, Cloud Computing, Artificial Intelligence, GPU Programming

American International University - Bangladesh September 2023

BSc. in Computer Science and Engineering (CSE) *Cumulative GPA: 3.99/4.00*

- Relevant coursework: Algorithms & Data Structures, OOP, Database, Networks, Web Technologies, Data Mining, Advanced Programming with Java & .NET, Software Quality & Testing, Artificial Intelligence

Professional Experience

Software Engineer March 2023 – July 2026

Brain Station 23 PLC, Dhaka, Bangladesh

- Lead a team of **6** engineers delivering software solutions for large-scale enterprise clients across government services and international insurance sectors in highly regulated domains.
- Built an AI Agent integrating LLMs APIs to automate insurance claim processing, achieving **40%** faster submission times.
- Developed a conversational AI assistant that personalizes customer onboarding by inferring preferences from natural language, reducing manual intake effort significantly.
- Automated approval workflows supporting **10,000+** daily approvals, boosting operational efficiency through concurrency control and duplicate prevention.
- Built an integrated wallet and bank API system processing over **$3 million** in monthly payments and streamlining automated fee collection.
- Migrated **15** years of relational data with full schema mapping to improve report accuracy and reliability by **70%.**
- Architected a polyglot persistence strategy using MongoDB and SQL Server, reducing query latency from **12s to 500ms.**
- Migrated a legacy application to the latest .NET with comprehensive unit tests, reducing bugs by **30%**.
- Automated CI/CD pipelines with Jenkins and integrated OAuth authentication for secure third-party API connectivity.

Research Assistant (Part-time) December 2024 – Present

ELITE Research Lab, NY, USA (Remote)

- Pursuing applied research in machine learning and AI, focusing on generative models, predictive modeling, and computer vision.
- Developing AI-driven systems and tools, including context-aware conversational agents for real-world applications.
- Mentoring students in LLM, AI, and machine learning research, and supporting the development of research tools and AI applications.

Skills

Core Competencies: Problem Solving, Algorithms & Data Structures, OOP, Agile, Scrum

Programming: C#, C++, SQL, Java, JavaScript

Frameworks & Libraries: .NET, xUnit, Angular, Nest Js, Next Js

AI & ML: OpenAI API, Gemini API, Claude, RAG, Vector Search, Pinecone, Embedding Models

Database: MSSQL, MongoDB, PostgreSQL

Architecture & Design: Clean Architecture, Microservice, Vertical Slice, MVC, REST, CQRS, SOLID, DRY, YAGNI

Software & Tools: Git, GitHub, CI/CD, Jenkins, Docker, Azure Repos, Amazon S3, Claude Code, Cursor

Projects

GradAgent.ai | *AI-Powered Graduate Admissions Assistant*

- Built an ASP.NET Core backend for an admissions chatbot, grounding LLM answers in MongoDB vector search over university and professor data.
- Implemented dynamic prompt construction and context injection to optimize LLM responses based on retrieved user profiles, university data, and application requirements..
- Developed a tool-calling LLM agent for personalized application strategies, with a swappable multi-provider AI layer.
- Integrated RAG pipelines over a curated knowledge base of university requirements, admission statistics, and program details to ground agent responses in accurate, up-to-date information.
- Engineered a sliding window mechanism for token allocation and rate limiting to control LLM costs across API .

Publications

- **M. J. Hasan**, S. H. Shifat, J. Matubber, R. Hossain, M. A. Rahman, B. M. T. Haque, M. J. Hossen. "An in-depth exploration of machine learning methods for mental health state detection: a systematic review and analysis." *Frontiers in Digital Health*, vol. 7, 2026.
- **M. J. Hasan**, A. Das, J. Matubber, S. H. Shifat, M. K. Morol. "Enhanced Classification of Anxiety, Depression, and Stress Levels: A Comparative Analysis of DASS21 Questionnaire Data Augmentation and Classification Algorithms." *3rd International Conference on Computing Advancements (ICCA 2024)*, ACM.
- M. A. H. Reaj, R. D. Gupta, M. Y. Rahat, N. Fahad, **M. J. Hasan**, T. H. Liew. "GANDiff-FR: Hybrid GAN–Diffusion Synthesis for Causal Bias Attribution in Face Recognition." Accepted at *International Conference on Computer Vision and Data Mining (ICCVDM 2025)*.

Certifications

- Google Data Analytics Professional Certificate – Google (Coursera)

Achievements

- Ranked 6th in the Intra AIUB Programming Contest Fall 2021–22
- Solved 600+ algorithmic challenges on LeetCode, Codeforces, HackerRank, and similar platforms
- Selected for the Dean's List honors for outstanding academic performance
- Finalist in an app development competition hosted by IEEE India in 2020
- Top 1% in class; received Summa Cum Laude for an exceptional CGPA

Languages

English: Fluent

Finnish: Learning, currently enrolled in Finnish for Foreigners 1, Metropolia University of Applied Sciences